

Grade 1 Ontario Curriculum Algebra

Balance Scale Stories

Unit Overview: Balance Scale Stories

Grade: Grade 1 | **Strand:** Algebra

Target Expectations: C2.1, C2.2, C2.3

Learning Goal: Children will identify quantities that change and quantities that stay the same; determine whether addition and subtraction expressions are equivalent; and find equivalent relationships for whole numbers up to 50.

Lesson	Lesson Title	Learning Goal
Day 1	Change vs Stay the Same	I can name things that change and things that stay the same in my life.
Day 2	Are They Equal?	I can decide if two addition or subtraction expressions are equal.
Day 3	Find the Missing Number	I can find the missing number that makes two expressions equal.
Day 4	Many Ways to Make 10 (and More)	I can find many equivalent expressions for the same whole number up to 50.
Day 5	Balance Story Capstone	I can tell a balance story about something in real life and write the equation.

Lesson 1: Change vs Stay the Same

Goal: Some quantities CHANGE (people in class, weather); others STAY THE SAME (10 fingers, 4 corners).

Expectation: C2.1

Learning Goal: "I can name things that change and things that stay the same in my life."

Materials: M3_change_constant_cards, M10_balance_poster

1. Minds On (12 mins): Bella's Sorting Game

How many children are in class today? How many sides on a square? Which one CHANGES? Bella opens with her refrain: 'Bella's question — does it CHANGE or STAY?' (Class echoes.)

Bring children to the carpet. Reveal Bella the Balance Beam Scientist puppet. 'Friends — meet Bella. Bella is a SCIENTIST. She studies which things change and which things stay the same in the world.'

2. Action (25 mins): Change or Stay-the-Same Stations

- Whole group: sort 6 more cards as a class.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: ask 'Are there any quantities that CHANGE only sometimes?'
- Discuss the why.
- Circulate to record formative evidence on the change-or-stay snapshot tracker.
- INVENT-AND-COMPARE: pairs find a quantity in the room nobody else thought of — name it and decide CHANGES or STAYS.

3. Consolidation (13 mins): Bella's Closing Sort

Discuss:

- What are three things in your life that change?
- What are three things that stay the same?
- Why does the number of fingers on one hand stay the same?
- Why does the number of children in class change?
- Bella's preview: tomorrow we meet the BALANCE SCALE (Lesson 2 — C2.2).

Exit: Bella's refrain — class chants three times together: "Bella's question — does it CHANGE or STAY?" with hand-signs (open palm = CHANGE, closed fist = STAY).

Lesson 2: Are They Equal?

Goal: Two expressions can be EQUAL (same amount) or NOT EQUAL.

Expectation: C2.2

Learning Goal: "I can decide if two addition or subtraction expressions are equal."

Materials: M1_balance_scale, M2_balance_drawings, M4_equivalence_cards, M5_two_colour_counters

1. Minds On (12 mins): Bella's Balance Test

Bella has $5+3$ on one side. Bobby has $4+4$ on the other. Will the balance be level or tipped?

Bring children to the carpet. Bella + Bobby puppets. 'Friends — Bella and Bobby have a real balance scale. They want to know — are $5+3$ and $4+4$ the SAME?'

2. Action (25 mins): Balance Stations

- Whole group: test 4 more pairs on the balance.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: include 2 'looks-different but equal' cases ($5+3$ vs $7+1$; $10-3$ vs $4+3$) and 2 'looks-similar but not equal' cases ($4+4$ vs $4+5$; $8-2$ vs $8-3$).
- Discuss the equal sign.
- Circulate to record formative evidence on the equivalence tracker.

3. Consolidation (13 mins): Bella's Equal-Sign Brag

Discuss:

- What does the $=$ sign MEAN?
- What does the $\neq$ sign mean?
- How does the balance scale show equal?
- Are $5+3$ and $4+4$ equal?

Exit: Each child writes ONE pair they tested today on a sticky note: '____ ____ ____' (with $=$ or $\neq$).

Lesson 3: Find the Missing Number

Goal: Knowing one side of the balance lets me find the missing number. $4 + ? = 7$ means 3.

Expectation: C2.2

Learning Goal: "I can find the missing number that makes two expressions equal."

Materials: M1_balance_scale, M2_balance_drawings, M5_two_colour_counters, M6_missing_number_cards

1. Minds On (10 mins): Bella's Mystery Box

Bella has 4 counters on one side. She has 7 counters on the other side. How many MORE does she need on the first side to balance? Bella opens with her refrain: 'Find what BALANCES — that's the mystery!' (Class echoes.)

Yesterday Bella proved $5+3 = 4+4$ with the balance (Day 2 — Are They Equal). The balance still showed EQUAL because both sides held the same total. Today the balance still shows equal — but ONE number is HIDDEN. Bella calls it the MYSTERY NUMBER.

2. Action (25 mins): Mystery-Number Stations

- Whole group: solve 5 more mystery numbers.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: include 2 problems where the unknown is on the LEFT ($? + 4 = 9$ instead of $5 + ? = 9$).
- Discuss strategies.
- Circulate to record formative evidence on the missing-number tracker.
- STRATEGY COMPARE: 'Pair A counted up 7,8,9,10; Pair B used $6+4=10$ from memory.

3. Consolidation (10 mins): Bella's Mystery Brag

Discuss:

- What's a missing number?
- What strategy do you use to find a missing number?
- Why does $4 + ?$
- Can the mystery be on either side of the equal sign?
- Tomorrow we ask: how many DIFFERENT mysteries balance to the same total? (Lesson 4 — C2.2 many-ways).

Exit: Bella's refrain — class chants in unison: "Find what BALANCES — that's the mystery!" Each child fills sentence frame on a sticky: 'My mystery is ___ + ___ = ___.'

Lesson 4: Many Ways to Make 10 (and More)

Goal: There are many ways to make the same amount. $10 = 5+5 = 4+6 = 8+2 = 0+10 = 11-1$.

Expectation: C2.3

Learning Goal: "I can find many equivalent expressions for the same whole number up to 50."

Materials: M1_balance_scale, M2_balance_drawings, M4_equivalence_cards, M5_two_colour_counters

1. Minds On (12 mins): Bella's 'Many Ways to Make 10' Challenge

Bella says $5+5$ makes 10. Bobby says $4+6$ makes 10. Who's right? Bella opens with the unit refrain: 'Same total, NEW way!' (Class echoes.)

Bring children to the carpet. Bella + Bobby puppets. 'Friends — Bella and Bobby are arguing. Bella says $5+5$ makes 10. Bobby says $4+6$ makes 10. Who's right?'

2. Action (25 mins): Equivalent-Hunt Stations

- Whole group: find 5+ expressions for 15.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: include 3-term expressions ($10+5+5 = 20$).
- Discuss patterns.
- Circulate to record formative evidence on the equivalent-expressions tracker.
- STRATEGY COMPARE: two pairs share their lists side-by-side and NAME the moves they used (commutative swap $6+4 \rightarrow 4+6$, near-doubles shift $5+5 \rightarrow 4+6$, subtraction inverse $10 = 11-1$).

3. Consolidation (13 mins): Bella's Equivalence Brag

Discuss:

- What does EQUIVALENT mean?
- How many ways did you find to make 10?
- Can subtraction expressions make 10?
- Why are there so many ways?

Exit: Bella's refrain — class chants together: "Same total, NEW way — that's EQUIVALENT!" Each child reads their sticky aloud: 'My new way: ____'.

Lesson 5: Balance Story Capstone

Goal: We can use balance ideas to tell stories about real life.

Expectation: C2.3, C2.1, C2.2

Learning Goal: "I can tell a balance story about something in real life and write the equation."

Materials: M1_balance_scale, M2_balance_drawings, M5_two_colour_counters, M8_story_template

1. Minds On (10 mins): Bella's Balance Story Showcase

Bella tells a story about her cookie jar. Today YOU will tell a balance story. Bella opens: 'Two sides — same amount?' Class chants: 'THAT'S a balance!'

Bring children to the carpet. Bella + Bobby puppets. 'Friends — today is the BIG day. We've learned about change vs stay-the-same, equal vs not-equal, missing numbers, and many ways to make the same total. Now we put it ALL together with BALANCE STORIES.'

2. Action (22 mins): Story-Writing Capstone

- Hand out Balance Story Templates.
- Children work for 10-12 minutes individually.
- Stretch challenge: ask 5+ children to include a CHANGE element.
- Gallery walk: children swap templates with a partner and read each other's stories.
- Whole-class share: 5-6 children share their story aloud.

3. Consolidation (13 mins): Closing Brag and Certificate Ceremony

Discuss:

- What's the most surprising balance story you heard today?
- Did anyone use subtraction in their story?
- What's one thing about balance you'll remember?
- What's something in your life that BALANCES?

Exit: Bella & Bobby unit closure refrain: Bella calls 'Two sides — same amount?' and the whole class chants back 'THAT'S a balance!' Repeat 3x.

Name: _____

Date: _____

Bella's Sorting Game

Learning Goal: I can name things that change and things that stay the same in my life.

Instructions for the Student:

1. Sort each card into CHANGE or STAY THE SAME.
2. Name 3 things in YOUR life that CHANGE and 3 things that STAY THE SAME.
3. Pick ONE thing that CHANGES from your list.

Part 3: Why?

Pick ONE thing that CHANGES from your list. Why does it change? Write a sentence.

[illegible]

Name: _____

Date: _____

Bella's Balance Test

Learning Goal: I can decide if two addition or subtraction expressions are equal.

Instructions for the Student:

1. Write = or $\neq$ between each pair.
2. Each balance shows expressions on each pan.
3. Write your own pair of expressions that are EQUAL.

Part 2: Balance Drawings

Each balance shows expressions on each pan. Write = or $\neq$ AND draw whether the balance is level or tipped.

Name: _____

Date: _____

Bella's Mystery Numbers

Learning Goal: I can find the missing number that makes two expressions equal.

Instructions for the Student:

1. Find the missing number in each addition equation.
2. Find the missing number in each subtraction equation.
3. Each balance shows one side counted.

Part 3: Balance Mysteries

Each balance shows one side counted. Write the equation and find the missing number.

Name: _____

Date: _____

Bella's Many Ways

Learning Goal: I can find many equivalent expressions for the same whole number up to 50.

Instructions for the Student:

1. Write 8 different ways to make 10.
2. Write 6 different ways to make 25.
3. Pick your own target between 30 and 50.

Part 3: My Target

Pick your own target between 30 and 50. Write 5 different ways.

Date: _____

Learning Goal: I can tell a balance story about something in real life and write the equation.

1. Title your story.
2. Draw the balance scale.
3. Write the equation that matches your story.

Title your story. Then write 1-2 sentences about it.

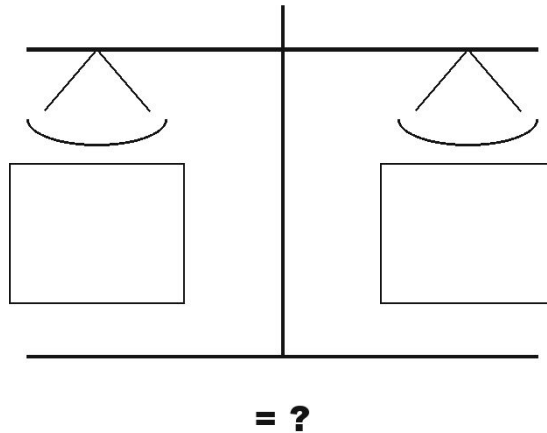
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Classroom Balance Scale

(M1_balance_scale)

Real two-pan balance scale (or cardboard prop) for class demonstrations.

Balance Mat



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 balance scale

Lessons: 2, 3, 5

Prep: ~12 min

TEACHER PREP

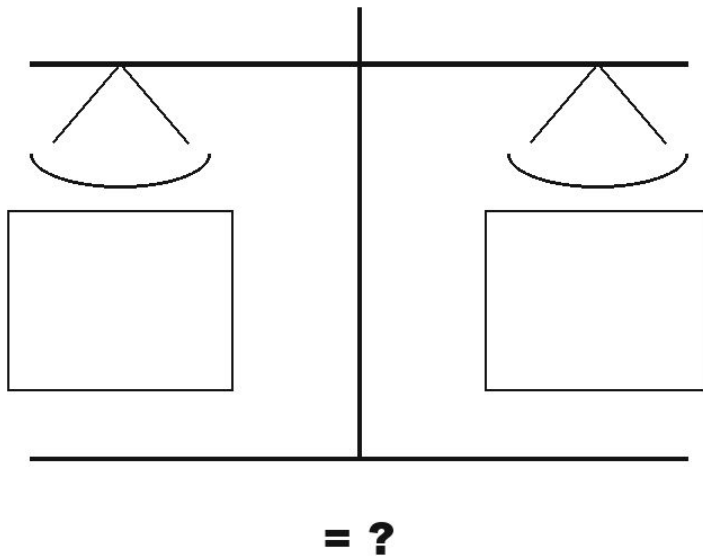
1. Source a real balance scale OR build cardboard version.
2. Print the setup sheet.
3. Prepare for whole-class demos.
4. Have ready Lessons 2-5.
5. Optional: extras.

Balance Scale Drawing Mat

(M2_balance_drawings)

Letter-size mat with a drawn balance scale for children to use.

Balance Mat



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 mat per child

Lessons: 2, 3, 4, 5

Prep: ~15 min

TEACHER PREP

1. Print 1 mat per child.

2. Laminate.

3. Hand out at start of Lesson 2.

4. Use through Lesson 5.

5. Optional: extras.

Change vs Stay-the-Same Cards

(M3_change_constant_cards)

24 picture cards for sorting in Lesson 1.

Change vs Stay-the-Same

GIVE 2 more	TAKE 1 away	Same group	ADD 3 more
MOVE to other side	Same total	REMOVE 4	Same balance

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 24 cards

Lessons: 1

Prep: ~22 min

TEACHER PREP

1. Print 4 pages on cardstock.

2. Laminate.

3. Cut into 24 cards.

4. Sort into per-pair sets.

5. Distribute Lesson 1.

Equivalence Cards

(M4_equivalence_cards)

30 cards with expression pairs for sorting equal/not equal in Lesson 2.

Expression Cards

$3 + 5$	$8 - 2$	$4 + 4$	$10 - 4$
$5 + 3$	$9 - 1$	$2 + 6$	$7 + 1$

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 5

Laminate: yes

USE IN CLASS

Quantity: 30 cards

Lessons: 2, 4

Prep: ~25 min

TEACHER PREP

1. Print 5 pages on cardstock.

2. Laminate.

3. Cut into 30 cards.

4. Sort by type.

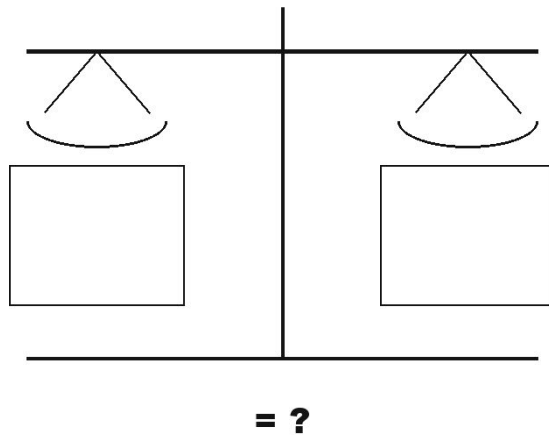
5. Distribute Lesson 2.

Two-Colour Counters

(M5_two_colour_counters)

20 per pair for balance modelling.

Balance Mat



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Confirm class set.

2. Print labels.

3. Sort into pair bags.

4. Have ready Lessons 2-5.

5. Optional.

USE IN CLASS

Quantity: 20 counters per pair

Lessons: 2, 3, 4, 5

Prep: ~8 min

Missing-Number Puzzle Cards

(M6_missing_number_cards)

24 cards for Lesson 3 missing-number puzzles.

Mystery Cards

$? + 4 = 9$	$? + 5 = 10$
$? + 6 = 12$	$? + 7 = 14$

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 24 cards

Lessons: 3

Prep: ~22 min

TEACHER PREP

1. Print 4 pages on cardstock.
2. Laminate.
3. Cut into 24 cards.
4. Sort by difficulty.
5. Distribute Lesson 3.

Equivalent-Target Cards

(M7_equivalent_target_cards)

12 'target number' cards for Lesson 4 hunt.

Target Cards

10	20	50	100
5	25	75	200

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 12 cards

Lessons: 4

Prep: ~12 min

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Distribute Lesson 4.

5. Optional: extras.

Balance Story Template

(M8_story_template)

Letter-size template for Lesson 5 capstone.

Story Template

Write your math story:



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~12 min

TEACHER PREP

1. Print 1 per child.
2. Hand out Lesson 5.
3. Children fill in.
4. Collect for assessment.
5. Display gallery.

Equal Sign Anchor Chart

(M9_equality_chart)

Large classroom anchor chart showing = and $\neq$.

Equation Table

Left	=	Right

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Lesson 2.

4. Reference Lessons 2-4.

5. Roll for re-use.

USE IN CLASS

Quantity: 1 chart

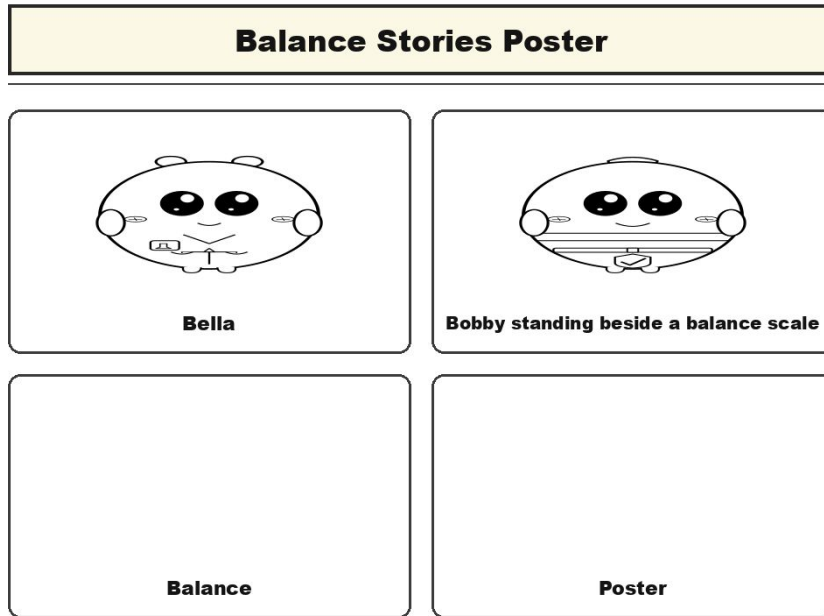
Lessons: 2, 3, 4

Prep: ~18 min

Balance Stories Poster

(M10_balance_poster)

Large classroom anchor showing Bella and Bobby with the balance scale.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

4. Reference all week.

5. Roll for re-use.

USE IN CLASS

Quantity: 1 poster

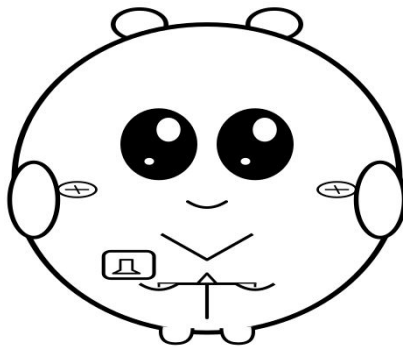
Lessons: 1, 2, 3, 4, 5

Prep: ~18 min

Bella

(char_bella_puppet)

Printable Bella illustration on a popsicle stick.



Bella the Balance Beam Scientist

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

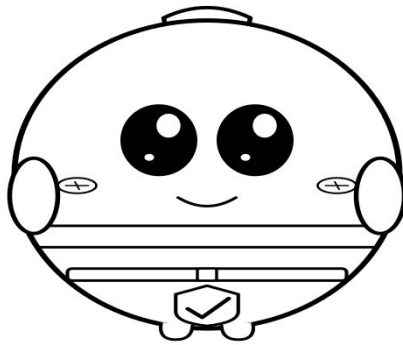
4. Optional: laminate.

5. Store flat.

Bobby the Balance Beam Buddy

(char_bobby_puppet)

Printable Bobby illustration on a popsicle stick.



Bobby the Balance Beam Buddy

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~10 min

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C2.1 identify quantities that can change and quantities that always remain the same in real-life contexts	Cannot distinguish change from constant. Treats today's count as fixed.	Sorts obvious cases with prompts.	Independently sorts most cases. Names own examples of each.	Articulates the WHY (definition vs experience). Generalises to new contexts.
C2.2 determine whether given pairs of addition and subtraction expressions are equivalent or not	Reads = as 'gives the answer'. Cannot test equivalence.	Tests with prompts. Sometimes treats expression-on-both -sides as wrong.	Independently tests pairs. Uses = and $\neq$ correctly. Solves missing-number problems.	Articulates = as 'same amount'. Reads bidirectionally.
C2.3 identify and use equivalent relationships for whole numbers up to 50, in various contexts	Lists 0-2 equivalent expressions for any number.	Lists 3-4 with prompts. Mostly uses addition only.	Lists 5+ varied expressions for numbers up to 50. Uses both + and -.	Lists 8+ varied including 3-term sums. Uses subtraction creatively.

Junior Balance Beam Scientist Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Balance Stories unit and demonstrating the skills of a Junior Balance Beam Scientist. Bella and Bobby congratulate the student.

By completing this unit, this student has demonstrated:

- Sorting quantities into change vs stay-the-same
- Testing pairs of expressions for equality
- Using $=$ and $\neq$ correctly
- Finding missing numbers in equations
- Generating equivalent expressions for any number to 50
- Telling a balance story with an equation

Marketplace Listing

A 5-day Ontario Grade-1 unit on Algebra (C2.1-C2.3) framed around 'Balance Scale Stories' — Bella the Balance Beam Scientist and Bobby her Balance Beam Buddy.

Price: \$9.99 CAD

Pages: ~35

Classroom time: 240 min

Teacher prep: ~90 min

Top tags: grade-1, math, ontario-curriculum, algebra, equivalence, missing-number, balance-scale, canadian

What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
- 5 student worksheets aligned to each lesson
- 1 mid-unit formative assessment (Equality Check-In)
- 10 printable teacher manipulatives plus 2 character puppets: classroom balance scale instructions, balance scale drawing mat, change-vs-constant cards (24),...
- 1 end-of-unit reflection sheet with 5 self-assessment prompts
- 4 class observation trackers: 1 diagnostic + 3 formative
- 1 three-row summative rubric covering C2.1-C2.3 across 4 levels
- 1 summative-administration guide tying the rubric to Lesson 5 worksheet
- Junior Balance Beam Scientist certificate (one per child)
- Detailed expanded walkthroughs for every lesson
- Differentiation tracks built into every lesson